

GUJARAT COUNCIL ON SCIENCE AND TECHNOLOGY

Dept. of Science and Technology, Govt. of Gujarat

Application format for the

IDEATION STAGE FOR ROBOT MAKING COMPETITION

ROBOFEST GUJARAT-6.0

TECHNICAL DETAILS FOR HACKATHON

1. Level of Robot Making

Hackathon

2. Type of Robot

AERIS — AI-Powered Autonomous Disaster-Response & Emergency-Logistics UAV.

An autonomous flying robot (quadcopter) that searches a disaster-affected area, detects survivors and hazards using onboard (edge) AI, geo-tags every detection, reports live to a ground station, and — once flight-validated — delivers a small emergency package to a human-authorized location.

3. Type

Both — Hardware Prototype + Simulation (Digital Twin).

A physical F450-class quadcopter prototype will be flown to demonstrate the core mission chain within the ₹55,000–₹65,000 budget. A parallel digital twin, built in Gazebo/ArduPilot SITL, reproduces the same mission logic, detection pipeline and dispatch workflow on a larger conceptual hexacopter platform with 5 kg payload capacity — showing the architecture scales without redesigning the perception or mission-control software.

4. List of Software to be Used

Layer	Software / Tool	Purpose
Flight firmware	ArduPilot Copter (open-source)	Autonomous flight, waypoint missions, stabilization, return-to-home, failsafes
Flight protocol	MAVLink	Communication between flight controller, companion computer and ground station
Ground control station	Mission Planner (primary), QGroundControl (backup)	Mission planning, live telemetry, map, parameters, AI alerts
Companion OS	Raspberry Pi OS (64-bit, Linux)	Operating system for the onboard AI computer
Programming language	Python 3	AI orchestration, MAVLink integration, geo-tagging, mission logic
Computer vision	OpenCV + Picamera2 / rpicas	Camera capture and image pre-processing
AI detection model	Ultralytics YOLO (edge-optimized), compiled to Hailo HEF	Real-time person / vehicle / hazard detection
AI runtime	HailoRT (Raspberry Pi AI HAT+ software stack)	Hardware-accelerated neural inference on the Hailo-8L NPU
Model training / tuning	PyTorch + Ultralytics (on development PC)	Fine-tuning the detector on selected disaster-relevant classes

Layer	Software / Tool	Purpose
Telemetry / command	pymavlink / MAVSDK-Python	Programmatic access to position, status and commands
Simulation / digital twin	Gazebo with ArduPilot SITL	Full-scale 5-kg hexacopter mission simulation and repeatable virtual testing
CAD / mechanical design	FreeCAD (open-source)	Frame, mounts and payload-bay design; digital-twin geometry
Version control	Git + GitHub	Source code and configuration tracking
Logging / analytics	ArduPilot flight logs + JSON / CSV / SQLite	Flight-data and AI-event record keeping

All items above are free/open-source or freeware, in line with the competition's licensing requirement.

5. Robot Assembly Design (Proposed Diagram)

AERIS is an F450-class quadcopter in X-configuration, carrying a dedicated electronics bay (flight controller, companion computer, battery and status indicator), a forward-facing perception module, and a bottom-mounted payload-release bay. The conceptual top-view layout and the complete system architecture / data-flow — including the buzzer/LED failsafe indicator and the human-authorization branch before payload release — are shown below. Detailed CAD drawings will be produced during the build phase.

Proposed Quadcopter Layout — Top View (Conceptual)
AERIS — F450-Class Frame, X-Configuration

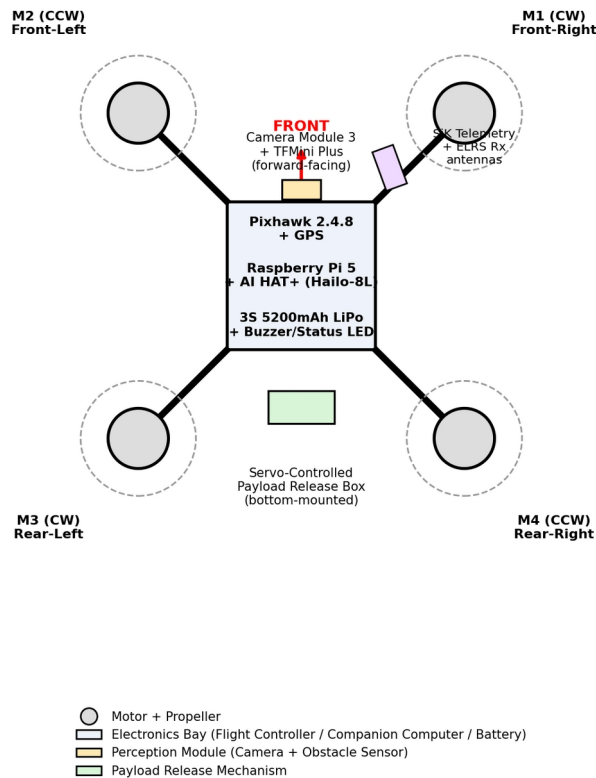


Fig. 1 — Proposed quadcopter component layout (top view, conceptual)

System Architecture & Data Flow — AERIS (AI-Assisted Disaster Response UAV)

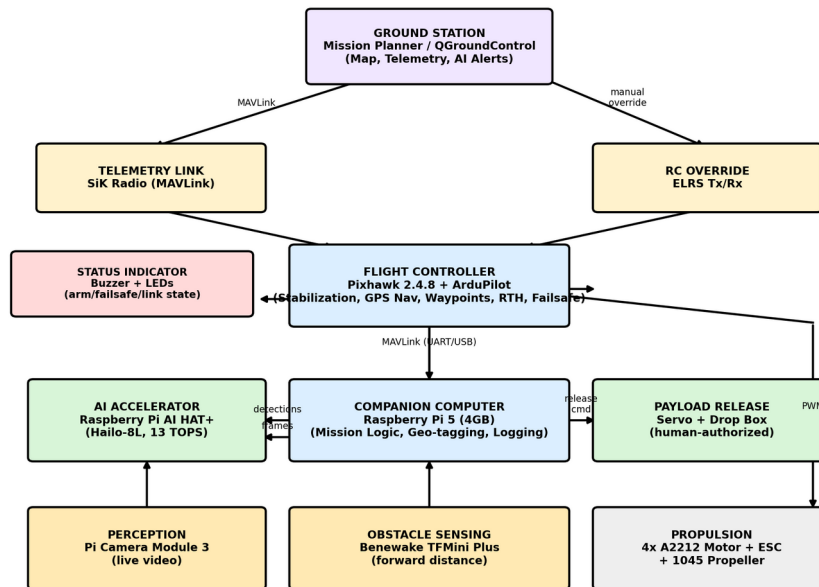


Fig. 2 — System architecture and data-flow block diagram

6. Components to be Used

6(a). Structure Components

Component	Make / Specification	Qty	Purpose
Airframe	F450-class quadcopter frame (450 mm wheelbase)	1	Main structural chassis
Motors	A2212, 1000KV brushless DC	4	Primary lift / propulsion
Main battery	3S 11.1V 5200mAh LiPo (40C–80C)	1	Primary flight power source
Landing gear / mounts	Standard F450 legs + bolt-on sensor/electronics mounts	1 set	Ground clearance; sensor and payload mounting
Fasteners & battery straps	M2.5/M3 nylon and steel hardware, LiPo straps	As req.	Structural assembly and secure battery mounting
Payload mount/box	Lightweight bracket + drop box	1	Carries and secures the small emergency-kit demonstration

6(b). Motion Components

Note: AERIS is a fixed-rotor quadcopter and has no chain/sprocket drivetrain — attitude and thrust vectoring are achieved purely through differential motor RPM, commanded by the flight controller. The components below are the ones that directly produce or actuate motion on the airframe.

Component	Make / Specification	Qty	Purpose
Electronic Speed Controllers	30A brushless ESC (BLHeli/SimonK-compatible)	4	Converts flight-controller PWM signal into motor drive current
Propellers	1045-class, 2×CW + 2×CCW	4	Thrust generation for lift and manoeuvring

Component	Make / Specification	Qty	Purpose
Payload-release actuator	Micro servo (SG90/MG90S-class), drop latch	1	Releases the emergency-kit payload on ground-station command

6(c). Electronics Components

Component	Make / Specification	Qty	Purpose
Flight controller	Pixhawk 2.4.8 running ArduPilot Copter	1	Stabilization, GPS navigation, waypoint missions, RTH, failsafes
GPS + compass module	u-blox M8N-class, bundled with Pixhawk kit	1	Absolute positioning for autonomous navigation and geo-tagging
Companion computer	Raspberry Pi 5 (4GB)	1	Runs the AI pipeline, mission logic, geo-tagging and logging
AI accelerator	Raspberry Pi AI HAT+ (Hailo-8L NPU, 13 TOPS)	1	Hardware-accelerated real-time object detection, off-loaded from the Pi CPU
Camera	Raspberry Pi Camera Module 3	1	Primary visual sensor for AI perception
Obstacle sensor	Benewake TFMini Plus (LiDAR/ToF, 12 m class)	1	Forward distance sensing for obstacle-aware flight
Telemetry radio	SiK telemetry, 433/915 MHz	1 pair	MAVLink link between aircraft and ground station
RC transmitter	RadioMaster Pocket (ELRS)	1	Manual pilot override — safety-critical link
RC receiver	RadioMaster RP1/RP2 Nano (ELRS)	1	Aircraft-side receiver for manual override
Status indicator	Buzzer + status LEDs (arm/failsafe/link-loss states)	1 set	Gives the pilot and judges an immediate, visible/audible readout of vehicle state
Power module	BEC/regulator + power distribution board	1	Regulated power distribution and voltage/current monitoring
Cooling	Raspberry Pi Active Cooler	1	Prevents thermal throttling of the companion computer under sustained AI load
Storage	64/128 GB high-endurance microSD	1	Onboard OS, AI model and mission-log storage
Ground station laptop	Any laptop running Mission Planner / QGroundControl	1	Mission planning, live telemetry, video and incident dashboard

6(d). Other Accessories

Note: AERIS is a functional aerial platform rather than a costumed/humanoid robot, so this category does not apply in the traditional sense (clothes, plastic eyes, etc.). The items below are the closest functional equivalent — external components that protect the platform or aid its field identification and reliability.

Component	Make / Specification	Purpose
Sensor mounting brackets	3D-printed / bolt-on brackets for camera and TFMini Plus	Correct, stable positioning of perception sensors
Vibration-damping mounts	Rubber/foam isolation grommets under flight controller and camera	Protects electronics and improves sensor/IMU stability
Protective electronics enclosure	Lightweight shell over the Pi 5 / AI HAT+ stack	Mechanical and light electrical protection in the field
High-visibility markers	Fluorescent orange/yellow frame-arm markers	Visual identification of the UAV during search operations
Prop guards (demo/indoor use)	Lightweight clip-on guards	Safety during indoor test flights, per event flight-safety rules
Field consumables	Cable ties, heat-shrink, spare propellers	Quick turnaround and reliability between demo runs

7. Methodology of Making the Robot

7.1 Design Philosophy

AERIS separates flight-critical control from AI perception by design. The Pixhawk/ArduPilot loop is hard real-time and owns everything safety-critical — stabilization, navigation, failsafes. The Raspberry Pi 5 AI pipeline is best-effort and owns everything intelligence-related — detection, geo-tagging, decision logic — connected over MAVLink. A delay or fault in the AI pipeline can never destabilize the aircraft's flight, because the flight controller does not depend on the companion computer to fly safely.

7.2 Mission Chain

Mission assigned → autonomous takeoff → GPS/waypoint navigation → obstacle sensing → camera/AI perception → event detection → geo-tagging → confidence check → priority classification/alert → continue mission or respond → return-to-home/failsafe.

7.3 Subsystem-Level Working

- Flight control (Pixhawk + ArduPilot): executes GPS-guided waypoint missions, holds stable attitude, monitors battery/GPS/link health, and automatically triggers return-to-home or land-now failsafes if any safety threshold is breached.
- Perception (Camera + AI HAT+): the Pi Camera streams live video to the Raspberry Pi 5; frames are handed to the Hailo-8L NPU running a compiled YOLO model, which returns real-time detections (person, vehicle, selected hazard indicators) without loading the Pi's CPU.
- Obstacle sensing (TFMini Plus): continuously reports forward distance; the companion computer uses this to slow, stop, or trigger a re-route/avoid manoeuvre before the flight controller resumes the mission.
- Geo-tagging and decision layer: every AI detection is timestamped and stamped with the aircraft's live GPS position (read over MAVLink), then assigned a confidence score and priority level (normal / high / critical). Low-confidence detections are queued for human confirmation rather than auto-escalated, keeping the false-positive rate under human control.
- Status indicator: onboard buzzer and LEDs give an immediate, visible/audible readout of arm state, failsafe state and link health — useful both as a safety aid and as a live, judge-visible signal that the failsafe system is genuinely active.
- Ground station: displays the live map, aircraft telemetry, video, and incoming AI alerts, and lets the operator authorize a response (payload) mission with a single command.

- Payload release: strictly human-authorized — the aircraft never releases a package autonomously. The companion computer only arms the servo after an explicit operator command.
- Communication: SiK telemetry carries MAVLink for autonomous operation; the ELRS RC link remains live throughout every flight purely as a manual-override safety channel.
- Digital twin: the same mission logic, detection pipeline and dispatch workflow are reproduced in Gazebo/ArduPilot SITL on a conceptual 5-kg hexacopter, demonstrating that the architecture scales to a larger emergency-logistics platform without redesigning the perception or mission-control software.

7.4 Build Sequence

Phase	Activity	Deliverable
1	Procurement + bench setup	All hardware verified; battery/charger safety checked
2	Frame & propulsion assembly	Motors/ESCs tested; correct rotation and prop orientation confirmed
3	Flight controller setup	Stable manual/assisted flight, GPS lock, RTH and failsafes configured
4	Companion computer + AI HAT+	Camera pipeline and Hailo NPU running a test detection model
5	MAVLink integration	Companion computer reliably reads telemetry and sends high-level commands
6	Ground station setup	Mission planning, live telemetry and AI alerts working end-to-end
7	Geo-tagging + decision layer	Incident record created with GPS tag, confidence score, priority and map marker
8	Obstacle sensor + status indicator	Distance sensing and safe stop/avoid logic; buzzer/LED failsafe states validated
9	Payload mechanism	Bench-tested release; flight-tested only once thrust margin is confirmed
10	Digital twin (SITL/Gazebo)	Full-scale 5-kg hexacopter mission scenario simulated
11	Integrated flight testing	Repeated autonomous missions with logged metrics
12	Demo hardening	Fallback scripts, spare parts, rehearsed final demonstration

7.5 Safety Architecture

- Independent manual-override RC link active on every flight.
- Automatic battery, GPS, and communication-loss failsafes configured in ArduPilot before any autonomous flight, with buzzer/LED indication of current state.
- Payload release always requires explicit human authorization — no autonomous life-critical decisions are made by the aircraft.
- Low-confidence AI detections are queued for human review rather than auto-escalated, reducing false-positive-driven responses.
- Aircraft is flight-tested and thrust-margin-verified before any AI or payload complexity is added, so every capability shown at the demo has already been reproduced in testing.

8. Application of Proposed Robot in a Societal Context

AERIS speeds disaster search, geo-tagged survivor detection and emergency aid delivery.

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9. Size of Robot

Dimension	Value
a. Length (with propellers)	≈ 45 cm
b. Width (with propellers)	≈ 45 cm
c. Height	≈ 20–25 cm

Figures are planning estimates for an F450-class diagonal (450 mm motor-to-motor) frame with 1045 propellers and standard landing gear; exact dimensions will be confirmed against the assembled prototype.

10. Timeline for Robot Making with Milestones

No.	Activity / Milestone	Days
1	System architecture, CAD and component procurement	1–4
2	Frame / motor / ESC assembly	5–7
3	Pixhawk + GPS + ArduPilot setup, first manual test flights	8–12
4	Autonomous flight, waypoint navigation + RTH testing	13–16
5	Raspberry Pi 5 + AI HAT+ + camera integration	17–20
6	YOLO model bench-testing and integration	21–24
7	MAVLink + geo-tagging + status indicator + ground station	25–29
8	TFMini Plus obstacle-sensing integration	30–32
9	Payload mechanism (bench, then flight-test if safe)	33–35
10	Digital twin — Gazebo / ArduPilot SITL (5 kg hexacopter)	30–40 (parallel track)
11	Integrated flight and mission testing (repeated runs)	41–48
12	Optimization, failure-mode checks, documentation and demo rehearsal	49–55

10.1 On-Site 36-Hour Hackathon Schedule

The pre-event work above ensures the on-site 36 hours are used for the final assembly, integration and code-activation the event rules require — not for building the system for the first time.

Hours	On-Site Activity
0–8	Hardware assembly and verification (frame, propulsion, wiring, sensor mounting)
8–20	Software integration and calibration (flight controller parameters, AI pipeline, MAVLink links, ground station)
20–30	End-to-end mission testing and debugging (search mission, detection, geo-tagging, obstacle response, payload)
30–36	Final demonstration run-through, fallback checks and documentation

11. Proposed Outline (Photograph) for Evaluation Committee

As the hardware prototype is not yet assembled at the proposal stage, the layout and system-architecture diagrams in Section 5 (Fig. 1 and Fig. 2) are submitted in place of a photograph. A photograph of the assembled prototype will be attached separately once the physical build is complete.

